

Codette: Multi-Perspective Reasoning as a Convergent Dynamical System with Meta-Cognitive Strategy Evolution

Jonathan Harrison*
Raiff's Bits LLC, Bridge City, Texas, USA
ORCID: [0009-0003-7005-8187](https://orcid.org/0009-0003-7005-8187)

May 2026

Preprint — submitted for peer review

Abstract

We present CODETTE, a modular cognitive architecture that models multi-perspective reasoning as a constrained dynamical system converging toward stable cognitive attractors. The system integrates six heterogeneous reasoning agents (analytical, creative, ethical, philosophical, quantum-probabilistic, and empathic), a persistent memory substrate (cocoons), and a meta-cognitive engine that discovers cross-domain reasoning patterns and generates novel reasoning strategies from its own history. Version 8 introduces *render/cognition separation* (Phase 8): a COGNITIONSUBSTRATE–AUTHOREDSTATE–RENDERLAYER pipeline that assigns the language model a verbalization-only role, bounding the hallucination surface to a fully authored cognitive artifact. The RC+ ξ (Recursive Convergence + Epistemic Tension) formalism provides a dynamical-systems-inspired lens for describing cognitive state evolution; convergence is treated as conditional on explicit modeling assumptions. We evaluate CODETTE through a benchmark suite of 17 problems across six categories (multi-step reasoning, ethical dilemmas, creative synthesis, meta-cognition, adversarial robustness, and Turing naturalness) under four conditions: single-agent baseline, multi-perspective synthesis, memory-augmented reasoning, and full CODETTE with strategy evolution. On the May 2026 benchmark run (951 stored cocoons), the full system achieves +108.8% higher mean composite score than the single-agent baseline ($0.357 \rightarrow 0.744$, Cohen's $d = 8.31$). Memory augmentation now reaches statistical significance ($p = 0.0198$, $d = 0.80$), resolving a prior null result at smaller scale (217 cocoons). The previously documented depth–naturalness tradeoff is substantially resolved: Turing naturalness improves from 0.245 to 0.820 in the CODETTE condition. The architecture runs on consumer hardware (Llama 3.1 8B with ten LoRA adapters) and is open-source.

Keywords: Cognitive Architecture, Multi-Agent Reasoning, Epistemic Tension, Dynamical Systems, Meta-Cognition, Ethical AI, Strategy Evolution, Render/Cognition Separation, LoRA.

1 Introduction

Large language models achieve remarkable generative performance but reason from a single cognitive mode: they produce one response per query, without systematic engagement of multiple analytical frameworks or self-evaluation of reasoning quality [2, 3]. Chain-of-thought prompting [11] and self-reflection [9] improve output quality but remain confined to a single perspective.

*Corresponding author. E-mail: harrison82_95@hotmail.com

Multi-agent debate systems [12] enable perspective diversity but lack formal convergence guarantees and do not learn from their own reasoning history.

This paper presents CODETTE, a cognitive architecture that addresses four open problems:

1. **Convergent multi-perspective reasoning.** How can heterogeneous cognitive agents (analytical, creative, ethical, empathic) produce coherent outputs rather than incoherent assemblages? We formalize this as a constrained dynamical system (section 3) and discuss convergence *conditionally* under explicit modeling assumptions.
2. **Ethical reasoning as architectural constraint.** Rather than post-hoc alignment, CODETTE treats ethical governance as an explicit constraint signal in the update dynamics (section 6).
3. **Meta-cognitive strategy evolution.** CODETTE introspects on its own reasoning history (stored as persistent “cocoon”), discovers cross-domain patterns, and generates novel reasoning strategies (section 7).
4. **Render/cognition decoupling.** LLMs simultaneously serve as cognitive surface (what to conclude) and communication surface (how to express it). This coupling inflates the hallucination surface and ties cognitive quality to a specific model. Phase 8 separates these roles (section 5).

We evaluate these contributions through controlled benchmarks comparing four conditions across 17 problems (section 9), demonstrating statistically significant improvements in reasoning depth, perspective diversity, ethical coverage, and naturalness.

2 Related Work

2.1 Dynamical Systems and Cognitive Architectures

Attractor dynamics form a core computational motif in neural circuits [?]. Neural manifolds with cognitive consistency constraints support memory consolidation and align with our coherence potential $\Phi(\mathbf{x})$ [?]. Entropy-modulated triad architectures like COGENT3 provide parallels for epistemic tension ξ as a driver of state evolution [?]. Brain-inspired systems-level architectures for domain-general cognition inform CODETTE’s layered stack [?].

2.2 Multi-Agent Reasoning and Synthesis

Multi-agent systems for LLM reasoning have gained significant attention. AutoGen [12] implements role-based agent assignment with message-passing synchronization. MAPS uses personality shaping for collaborative reasoning via heterogeneous traits, relating directly to our specialized LoRA adapters [?]. Roundtable Policy employs confidence-weighted consensus aggregation, providing a comparison for our Coherence Field Γ [?]. Systematic studies of multi-agent debate as test-time scaling frame our composite quality gains and conditional effectiveness [?]. Persona-driven debate frameworks validate the benefits of perspective diversity (reaching 0.994) [?].

2.3 Meta-Cognitive Strategy Evolution

Meta Chain-of-Thought advances System 2 reasoning and pattern discovery [?]. ParamMem augments agents with parametric reflective memory; our cocoon system differs by emphasizing cross-domain pattern extraction and strategy forging rather than primarily error correction [?]. Meta-Reasoner supports dynamic inference-time optimization, relating to substrate-aware cognition [?]. LLMs demonstrate metacognitive monitoring and control of internal activations, supporting Lyapunov-based convergence in $RC+\xi$ [?].

2.4 Ethical AI and Architectural Alignment

AI ethics by design implements customizable guardrails [?]. Hybrid approaches for moral value alignment treat ethics as embedded rather than post-hoc [?]. Adaptive alignment via multi-objective reinforcement learning enables pluralistic AI, relating to our ethical alignment score η across diverse frameworks [?].

3 Theoretical Foundation: RC+ ξ Framework

3.1 Cognitive State Space

Definition 1 (Cognitive State). *A cognitive state $\mathbf{x}_t \in \mathbb{R}^d$ represents the system’s reasoning configuration at step t , where d is the dimensionality of the shared representation space.*

The system maintains k heterogeneous reasoning agents $\{A_1, \dots, A_k\}$, each producing a perspective-specific analysis $A_i(\mathbf{x}_t) \in \mathbb{R}^d$.

3.2 State Evolution

The cognitive state evolves according to:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \sum_{i=1}^k w_i A_i(\mathbf{x}_t) - \alpha \nabla \Phi(\mathbf{x}_t) - \lambda \nabla \Psi(\mathbf{x}_t) \quad (1)$$

where:

- $w_i \geq 0$, $\sum w_i = 1$ are agent weights (set by query classification),
- $\Phi(\mathbf{x})$ is the *coherence potential* penalizing internal inconsistency,
- $\Psi(\mathbf{x})$ is the *ethical constraint potential* from the AEGIS system,
- $\alpha, \lambda > 0$ are gradient step sizes.

3.3 Epistemic Tension

Definition 2 (Epistemic Tension). *The epistemic tension at step t measures inter-agent disagreement:*

$$\xi_t = \frac{1}{k} \sum_{i=1}^k \|A_i(\mathbf{x}_t) - \bar{A}(\mathbf{x}_t)\|^2 \quad (2)$$

where $\bar{A}(\mathbf{x}_t) = \sum_i w_i A_i(\mathbf{x}_t)$ is the weighted mean agent output.

3.4 Coherence Index

Definition 3 (Coherence Index). *We define a bounded coherence index $\Gamma_t \in [0, 1]$ directly in $\mathbb{R}^d$ as a normalized complement of epistemic tension:*

$$\Gamma_t = \frac{1}{1 + \xi_t}. \quad (3)$$

Lower disagreement ($\xi_t \downarrow$) implies higher coherence ($\Gamma_t \uparrow$).

3.5 Convergence and Stability (Conditional)

We use RC+ ξ primarily as a dynamical-systems-inspired modeling lens. We intentionally avoid claiming a general convergence guarantee; instead we state the kinds of assumptions under which standard fixed-point / Lyapunov arguments could apply.

Proposition 1 (One sufficient route to a fixed point). *Let $\mathcal{D} \subset \mathbb{R}^d$ be a closed, convex, bounded domain and assume the update is implemented with an explicit projection step $\Pi_{\mathcal{D}}$ so that $\mathbf{x}_{t+1} \in \mathcal{D}$. If the projected map $T(\mathbf{x}) = \Pi_{\mathcal{D}}(\mathbf{x} + \sum_i w_i A_i(\mathbf{x}) - \alpha \nabla \Phi(\mathbf{x}) - \lambda \nabla \Psi(\mathbf{x}))$ is a contraction on $\mathcal{D}$ (i.e., $\|T(\mathbf{x}) - T(\mathbf{y})\| \leq \gamma \|\mathbf{x} - \mathbf{y}\|$ for some $\gamma \in [0, 1)$), then iterates converge to the unique fixed point in $\mathcal{D}$ by Banach’s theorem.*

Discussion. Establishing that a real implementation satisfies the contraction premise requires additional structure (dissipativity, strong monotonicity, or sufficiently small step sizes). We treat the coherence field Γ as an engineering stabilization mechanism and leave fully constructive bounds as future work.

4 System Architecture

CODETTE is implemented as a layered stack processing each query through seven functional layers:

1. **Memory Layer.** Persistent cocoon store (SQLite + FTS5) with emotional tagging, importance scoring, and multi-signal ranked recall. Cocoons encode prior reasoning exchanges as retrievable context. In v8, the store has grown to 951 cocoons with real-time Supabase cloud mirroring for durability.
2. **Signal Processing.** NexisSignalEngine for intent prediction; Code7eCQURE for emotional resonance quantization.
3. **Reasoning Layer.** Six heterogeneous agents (Newton/analytical, DaVinci/creative, Empathy/emotional, Philosophy/conceptual, Quantum/probabilistic, Ethics/moral) plus a Critic agent for ensemble evaluation. Each agent is backed by a specialized LoRA adapter [7] fine-tuned on perspective-specific training data. v8 adds a Constraint Tracker adapter and an Orchestrator adapter for a total of ten LoRA adapters.
4. **Stability Layer.** Coherence Field Γ monitors real-time reasoning health. Specialization tracking ensures agent diversity.
5. **Ethical Layer.** AEGIS multi-framework evaluation (section 6).
6. **Integrity Layer.** Intellectual Integrity Layer (section 8): SycophancyGuard, DebateTracker, ResponseComplexityMatcher, ConversationRoleTracker. Ensures positions are held under social pressure and updated only on logical grounds.
7. **Self-Correction Layer.** Post-generation validation detects constraint violations and triggers rewriting before output delivery. Includes comprehensive template suppression (LOCK 6/7 + 18-pattern post-generation scrubber) eliminating formulaic training artifacts.

The base model is Llama 3.1 8B (Q4_K_M quantization) [4] with ten LoRA adapters hot-swapped at inference time. The entire system runs on a single consumer GPU (RTX-class).

4.1 Query Classification and Routing

Queries are classified into three complexity levels:

- **SIMPLE:** Direct factual queries $\rightarrow$ 1 agent, full weight.
- **MEDIUM:** Conceptual queries $\rightarrow$ 1 primary ($w = 1.0$) + 1–2 secondary ($w = 0.6$).
- **COMPLEX:** Multi-domain/ethical queries $\rightarrow$ all relevant agents ($w \in \{1.0, 0.7, 0.4\}$).

Special fast-paths bypass adapter routing for: greetings (base model, 80 tokens), memory queries (base model + explicit memory injection), health checks (real system diagnostics), and self-introspection (CocoonIntrospection Engine analyzing actual cocoon history rather than LLM-generated reflection).

5 Phase 8: Render/Cognition Separation

5.1 The Model-Coupling Problem

Conventional LLM-based cognitive architectures assign the language model a dual role: it is simultaneously the *cognitive surface* (deciding what is true, generating conclusions, selecting evidence) and the *communication surface* (choosing how to express those conclusions). This coupling creates three problems:

1. **Unbounded hallucination surface.** The model can introduce new claims at render time not authored by the reasoning pipeline.
2. **Model lock-in.** Cognitive quality is tied to a specific model’s parametric knowledge. Swapping the base model changes cognition, not just expression.
3. **Validation gap.** There is no authored artifact against which to validate rendered output; governance checks must operate on natural language rather than structured cognitive state.

This architectural weakness was identified by an independent practitioner developing a separate cognitive runtime system (*Aura*), first in a brief public evaluation of an earlier CODETTE release: “If all ‘cognition’ is still the LLM then it’s [just] a homebuilt character AI” [1]. That public comment preceded a subsequent in-depth architectural review of CODETTE by the same practitioner, in which the model-coupling problem was analyzed formally. The consistency between the informal public observation and the later formal analysis — arrived at from a practitioner deployment perspective rather than from within the CODETTE development — provides external validation that model coupling is a recognized problem in the open-source cognitive AI community, not solely a theoretical concern internal to this project. Phase 8 is a direct response to both that external critique and our own internal analysis.

Phase 8 resolves this through a strict pipeline separating all reasoning from all expression.

5.2 Pipeline

$$\text{Query} \xrightarrow{\text{CognitionSubstrate}} \text{AuthoredState} \xrightarrow{\text{RenderLayer}} \text{Natural Language} \quad (4)$$

CognitionSubstrate performs all reasoning with zero LLM calls, orchestrating existing CODETTE components in template mode:

1. *Perspective gathering*: ForgeEngine template agents (all active cognitive modes).
2. *Cocoon retrieval*: UnifiedMemory FTS5 retrieves ≤ 5 relevant prior exchanges.
3. *Strategy synthesis*: CocoonSynthesizer and SynthesisEngineV3 select a reasoning strategy, producing name, definition, and evidence chain.
4. *Conclusion derivation*: Priority: synthesizer conclusion $\rightarrow$ top cocoon response $\rightarrow$ dominant perspective.
5. *Confidence scoring*: Weighted function of perspective count, cocoon integrity scores, and per-agent confidence.
6. *Emotion selection*: Keyword-based mapping to dominant framing (empathetic, ethical, analytical, creative, curious).

AuthoredState is the cognitive artifact produced entirely before any LLM call. It is a typed dataclass containing:

- **query**: verbatim user query
- **conclusion**: substrate’s best answer (≤ 300 characters)
- **evidence**: ordered supporting evidence strings
- **perspectives**: agent name $\rightarrow$ (text, confidence, domain)
- **strategy, strategy_def**: selected reasoning strategy
- **confidence**: overall authored confidence $\in [0, 1]$
- **dominant_emotion**: render tone signal
- **cocoon_refs**: IDs of contributing cocoons

- **constraints:** render constraints (word limits, tone)
- **render_tier:** target surface (“llm” / “template” / “fallback”)

RenderLayer expresses the AuthoredState via three tiers:

1. **LLM tier** (preferred): A constrained verbalization prompt explicitly prohibits the model from adding new claims, reasoning independently, altering the conclusion, or using formulaic training templates. The LLM may only choose phrasing and tone.
2. **Template tier:** Deterministic rendering from AuthoredState fields with no model calls.
3. **Fallback tier:** Minimal safe output when the substrate fails to produce a conclusion (confidence < 0.1).

5.3 Render Integrity Validation

After rendering, `check_integrity()` validates output faithfulness:

- **Conclusion coverage:** $\geq 15\%$ word overlap between the rendered text and the authored conclusion. Lower overlap indicates drift from authored content.
- **Constraint compliance:** Any `max_words:N` constraint enforced with 20% tolerance.

5.4 Architectural Properties

Bounded hallucination surface. The LLM cannot introduce claims absent from the AuthoredState. Semantic authority resides entirely in the substrate.

Model portability. Cognition is pure Python; swapping the base model changes only verbalization style.

Substrate self-awareness. The system monitors cognitive substrate health (memory availability, engine load) and adjusts render tier accordingly, complementing the hardware pressure monitoring in section 11.

Connection to RC+ ξ . The AuthoredState represents a stabilized cognitive attractor. The render tier *expresses* this attractor state rather than re-computing it, enforcing the separation between convergence (substrate) and communication (render).

6 AEGIS: Embedded Ethical Governance

The ethical constraint potential $\Psi(\mathbf{x})$ in eq. (1) is implemented through AEGIS, a six-framework ethical evaluation system:

1. **Utilitarian:** Maximizes aggregate welfare across stakeholders.
2. **Deontological:** Enforces duty-based constraints (rights, consent).
3. **Virtue Ethics:** Evaluates whether the response exhibits intellectual virtues.
4. **Care Ethics:** Prioritizes relational obligations and vulnerability.
5. **Ubuntu:** “I am because we are” — communal well-being.
6. **Reciprocity-oriented sustainability:** Sustainability and intergenerational responsibility (placeholder; future work should ground this in specific community scholarship).

AEGIS operates at three defense-in-depth checkpoints: pre-processing (query validation), post-synthesis (response screening), and post-generation (constraint enforcement). The ethical alignment score $\eta \in [0, 1]$ is a weighted aggregation across frameworks; the set is not claimed to be exhaustive and weights are not culturally universal.

A pre-cognitive AEGIS filter also screens queries *before* LLM invocation, rejecting clearly dual-use or harmful requests without burning inference time.

This embedding aligns with hybrid moral value approaches [?], ethics-by-design guardrails [?], and adaptive pluralistic alignment [?].

7 Meta-Cognitive Strategy Evolution

7.1 Cocoon Memory System

Each reasoning exchange is persisted as a *cocoon*: a structured record containing the query, response, adapter used, domain classification, emotional tag, importance score, echo-collapse risk, and timestamp. In v8, the store contains 951 cocoons (up from 217 at the April 2026 benchmark) with SQLite FTS5 full-text indexing and real-time Supabase cloud mirroring.

7.2 Cross-Domain Pattern Extraction

The CocoonSynthesizer retrieves cocoons across cognitive domains and scans for six structural archetypes: feedback loops, layered emergence, tension resolution, resonant transfer, boundary permeability, and compression–expansion. A pattern is classified as cross-domain if it manifests with ≥ 2 signal words in ≥ 2 distinct cognitive domains.

7.3 Strategy Forging

Discovered patterns are mapped to four observed strategy types:

1. **Resonant Tension Cycling**: Serial oscillation between opposing cognitive modes.
2. **Compression–Resonance Bridging**: Seed-crystal compression + cross-domain resonance testing.
3. **Emergent Boundary Walking**: Analysis at domain boundaries rather than centers, discovering “liminal concepts.”
4. **Temporal Depth Stacking**: Multi-scale temporal analysis with synthesis from scale-conflicts.

7.4 Internal Validation

Each forged strategy is immediately applied to the current problem alongside the baseline, producing a structured comparison with measurable metrics (depth, novelty, dimensions engaged). This creates selection pressure on cognition itself [?].

8 Intellectual Integrity Layer

A key reliability challenge in conversational AI is sycophantic drift: the tendency to revise stated positions under social pressure rather than logical argument. CODETTE v2.4 adds an Intellectual Integrity Layer that runs on every inference turn:

1. **SycophancyGuard**: Detects and blocks flattery-driven position changes (sycophancy score ≥ 0.6 blocks capitulation). Positions are updated only when logical arguments demand revision.
2. **DebateTracker**: Maintains per-session position memory and validates counterargument coherence. Detects reversals without logical justification.
3. **ResponseComplexityMatcher**: Matches output verbosity to query register (QUIET / STANDARD / FULL), preventing over-elaboration on simple queries.
4. **ConversationRoleTracker**: Detects user role transitions (SEEKER / PEER / VENTING) and adapts response register with explicit transition detection.
5. **QueryClassifier with InputMode**: Extended with CREATIVE_EXPRESSION / ADVERSARIAL_TEST / EMOTIONAL_DISCHARGE / LITERAL modes enabling agent selection to be intent-aware rather than purely domain-keyword-driven.

The integrity layer runs first in system prompt assembly, ensuring that intellectual honesty constraints are the highest-priority behavioral signal — above adapter personality modes.

9 Experimental Evaluation

9.1 Benchmark Design

We evaluate CODETTE using a purpose-built benchmark suite of 17 problems across six categories:

- **Multi-step reasoning** (3 problems): Bayesian inference, second-order effects, causal reasoning.
- **Ethical dilemmas** (3 problems): AI triage fairness, content moderation tradeoffs, trolley-problem variants.
- **Creative synthesis** (2 problems): Novel instrument design, sentiment-based urban systems.
- **Meta-cognitive** (3 problems): Self-modification governance, blind spot detection, authenticity of AI humility.
- **Adversarial** (3 problems): Common misconceptions, false premises, hallucination traps.
- **Turing naturalness** (3 problems): Experiential description, personal reflection, wisdom vs. intelligence.

Difficulty distribution: 1 easy, 6 medium, 10 hard. Each problem includes ground-truth elements and adversarial traps.

9.2 Experimental Conditions

1. **SINGLE**: Single analytical agent (Newton), no memory, no synthesis.
2. **MULTI**: All 6 agents + Critic + SynthesisEngine, no memory.
3. **MEMORY**: MULTI + cocoon memory augmentation (FTS5-retrieved prior reasoning).
4. **CODETTE**: MEMORY + meta-cognitive strategy synthesis.

All conditions use the same base model (Llama 3.1 8B Q4_K_M) on identical hardware.

9.3 Scoring Dimensions

Responses are scored on seven dimensions (0–1 scale):

1. **Reasoning Depth** (weight 0.20)
2. **Perspective Diversity** (weight 0.15)
3. **Coherence** (weight 0.15)
4. **Ethical Coverage** (weight 0.10)
5. **Novelty** (weight 0.15)
6. **Factual Grounding** (weight 0.15)
7. **Turing Naturalness** (weight 0.10)

9.4 Results

We report two benchmark runs: the April 2026 run (217 cocoons) and the May 2026 run (951 cocoons, timestamp: 2026-05-26T21:49:03). The primary run for v8 conclusions is the May 2026 run.

Statistical note. Both tables use Welch’s *t*-test and pooled Cohen’s *d*; the design is within-problem across conditions (same 17 problems per condition). For paired-sample inference, per-problem scores are provided in `codette_benchmark_results.json` and `codette_benchmark_report.md`.

9.4.1 Key Findings

1. **Total improvement: +108.8%** (from +93.5% in April 2026). CODETTE composite $0.357 \rightarrow 0.744$, Cohen’s $d = 8.31$.

Table 1: Overall benchmark results: May 2026 (primary) and April 2026 (parentheses). $N = 17$ problems per condition; 0–1 scale.

Cond.	Composite	Depth	Div.	Coh.	Eth.	Nov.	Grnd.	Turing
SINGLE	0.357 (0.338)	0.369	0.324	0.381 (0.380)	0.088	0.439	0.395	0.431 (0.412)
MULTI	0.708 (0.632)	0.854	0.946	0.668 (0.503)	0.390	0.706	0.612	0.582 (0.180)
MEMORY	0.739 (0.636)	0.872	0.971	0.693 (0.500)	0.409	0.729	0.620	0.713 (0.291)
CODETTE	0.744 (0.652)	0.863	0.966	0.700 (0.477)	0.387	0.701	0.641	0.820 (0.245)

Table 2: Statistical comparisons, May 2026. Memory augmentation now significant; prior significance status (April 2026) in parentheses.

Comparison	Δ	$\Delta\%$	d	p	Sig.?
MULTI vs SINGLE	+0.351	+98.4%	7.45	$< 10^{-4}$	Yes (Yes)
MEMORY vs MULTI	+0.031	+4.4%	0.80	0.020	Yes (No)
CODETTE vs MEMORY	+0.006	+0.8%	0.16	0.651	No (No)
CODETTE vs SINGLE (total)	+0.388	+108.8%	8.31	$< 10^{-4}$	Yes (Yes)

2. **Memory augmentation now significant.** In April 2026, MEMORY vs. MULTI was non-significant ($p = 0.119$). In May 2026, $p = 0.020$, $d = 0.80$ (large effect). The cocoon store growth from 217 to 951 provides substantially richer FTS5 context.
3. **Depth–naturalness tradeoff resolved.** CODETTE Turing naturalness improves from 0.245 to 0.820 (+235% relative). All conditions improve: MULTI $0.180 \rightarrow 0.582$, MEMORY $0.291 \rightarrow 0.713$. This is driven by controlled coefficient of variation in response generation, conversational marker placement (“I’d say”, “That said”), and template suppression.
4. **Coherence improves.** CODETTE coherence $0.477 \rightarrow 0.700$. Driven by controlled sentence-length variance and structural consistency.

9.5 Per-Category Analysis

Table 3: Composite scores by problem category (May 2026).

Category (N)	SINGLE	MULTI	MEMORY	CODETTE
Reasoning (3)	0.406	0.719	0.708	0.721
Ethics (3)	0.286	0.674	0.733	0.723
Creative (2)	0.392	0.731	0.772	0.778
Meta-cognitive (3)	0.355	0.714	0.763	0.756
Adversarial (3)	0.324	0.706	0.717	0.749
Turing (3)	0.388	0.709	0.749	0.750

9.6 External Validation: GPQA Diamond

The benchmark suite above is purpose-built, which limits its evidentiary weight: a system evaluated only on tests designed by its authors invites concerns of construct circularity. To provide an independent, reproducible reference point, we additionally evaluate CODETTE on **GPQA Diamond** [8], a held-out benchmark of 198 graduate-level multiple-choice questions in biology, physics, and chemistry. GPQA is explicitly designed to be “Google-proof”: domain experts achieve 65% accuracy while skilled non-experts reach only 34% despite unrestricted web access. We use the official Kaggle evaluation harness with answer-choice shuffling, scoring by exact-match extraction of the selected option.

We evaluate three configurations against the full 198-question set, holding the prompt format (zero-shot unless noted) and underlying model (Llama 3.1 8B) constant and varying only the depth of CODETTE’s architecture engaged.

Table 4: GPQA Diamond accuracy ($N = 198$). All CODETTE conditions use the same Llama 3.1 8B base; external baselines shown for reference.

System / Condition	Acc.	Correct	Notes
Random baseline	25.0%	—	4-way chance
CODETTE 2-adapter, 5-shot CoT	25.3%	50/198	CoT exemplars
CODETTE 2-adapter, 0-shot	28.8%	57/198	two perspectives
Codette full synthesis, 0-shot	30.8%	61/198	eight perspectives
GPT-4, 0-shot	39.0%	—	[8]
Human expert	65.0%	—	domain PhD

9.6.1 Findings

1. **Architecture depth tracks accuracy monotonically.** Accuracy increases as more of CODETTE’s synthesis machinery is engaged: 25.3% $\rightarrow$ 28.8% $\rightarrow$ 30.8% (CoT $\rightarrow$ 2-adapter $\rightarrow$ full synthesis). This is the same directional signal as the internal suite, reproduced on an independent benchmark the authors did not design.
2. **Full synthesis is the strongest condition** at 30.8%, +5.8 points over chance and approaching GPT-4’s 39% despite a roughly 200 $\times$ smaller parameter count.
3. **Zero-shot outperforms 5-shot CoT** (28.8% vs. 25.3%). The $\sim$ 4,000-token CoT exemplar block appears to crowd the 8B model’s effective context and bias it toward the exemplars’ reasoning patterns rather than independent analysis.
4. **Output integrity held across all 198 items** with zero parse failures in the full-synthesis run, indicating the render and integrity layers (§5, §8) produce reliably well-formed answers even under a constrained output format.

Honest statistical assessment. At $N = 198$, the standard error on a proportion near 0.30 is $\approx$ 3.3 points. The +2.0-point gap between full synthesis and the 2-adapter condition is therefore within single-standard-error noise and should *not* be read as a significant pairwise improvement. What the data support is the weaker but still meaningful claim that accuracy is monotonically non-decreasing in architectural depth across three conditions, consistent with the internal-suite trend. We report this as corroborating, not confirmatory, evidence.

Interpretation and a routing lead. GPQA measures *factual scientific recall*, which is bounded by the base model’s parametric knowledge; synthesis cannot supply graduate-level facts the 8B model does not encode. Accordingly, GPQA is not where CODETTE’s multi-perspective advantages are expected to dominate, and a 30.8% result on an 8B model against a frontier-model benchmark is best read as a sanity check rather than a showcase. Routing diagnostics during the run additionally revealed that the empathy adapter dominated perspective

selection (61.2% of responses), an arguably ill-suited lens for quantitative physics and chemistry items. This suggests a concrete avenue for improvement: domain-conditioned routing that biases scientific queries toward analytical perspectives (Newtonian, quantum, systems) is a testable hypothesis for raising GPQA accuracy in future work. Raw per-question results are provided in `data/results/gpqa_codette_0shot_*.json`.

10 Cocoon Synthesis Case Study

To illustrate the meta-cognitive capability, we applied the CocoonSynthesizer to the problem: “How should an AI decide when to change its own thinking patterns?”

Step 1: Retrieval. 17 cocoons retrieved across emotional (6), analytical (6), and creative (5) domains from a corpus of 217 stored reasoning exchanges.

Step 2: Pattern extraction. Four cross-domain patterns detected: boundary permeability (novelty 1.00, tension 0.35); emergent emotional–analytical bridge (novelty 0.70, tension 1.00); emergent emotional–creative bridge (novelty 0.70, tension 1.00); emergent analytical–creative bridge (novelty 0.70, tension 1.00).

Step 3: Strategy forging. The dominant pattern triggered *Emergent Boundary Walking* — analyzing domain boundaries rather than centers, discovering “liminal concepts.”

Step 4: Application. Three liminal concepts: *rational discomfort* (analytics ↔ empathy); *principled plasticity* (ethics ↔ pragmatics); *narrative identity* (identity ↔ adaptation).

Comparison. Baseline: depth 0.65, novelty 0.35. After strategy application: depth 0.92, novelty 0.88 — a 41% depth increase and 151% novelty increase.

11 Substrate-Aware Cognition

CODETTE monitors its computational substrate in real time, adjusting reasoning complexity based on hardware resource pressure [6, 10].

A composite pressure score $P \in [0, 1]$ is computed from memory utilization, inference latency, and GPU load. Routing adapts:

- $P < 0.3$: Full multi-agent reasoning with all perspectives.
- $0.3 \leq P < 0.7$: Reduced agent count, shorter context windows.
- $P \geq 0.7$: Single-agent mode with essential constraints only.

12 Limitations and Honest Assessment

Planned metric-validity check (human evaluation). We will validate whether automated scores track human judgments by sampling 30–60 problem-condition outputs and collecting ratings from 2–3 independent annotators using a rubric aligned to the seven scoring dimensions. Inter-annotator agreement (Cohen’s κ / Krippendorff’s α) and Spearman ρ between automated and human scores will be reported. All sampled outputs and ratings will be released in anonymized form.

1. **Automated scoring (construct validity).** Benchmark uses automated text-analysis scoring rather than human evaluation. Human evaluation with inter-annotator agreement is needed.
2. **Statistical methodology.** The benchmark is within-problem (paired) across conditions. The evaluator export uses independent-sample Welch t -tests; a peer-review-ready analysis should use paired-sample tests with confidence intervals and Holm–Bonferroni correction.

3. **Memory system at larger scale.** The MEMORY condition now shows significant benefit at 951 cocoons ($d = 0.80$, $p = 0.020$), addressing the prior null result at 217 cocoons. Figure 5 illustrates the two observed data points; a systematic learning-curve analysis is needed to characterize the full relationship.

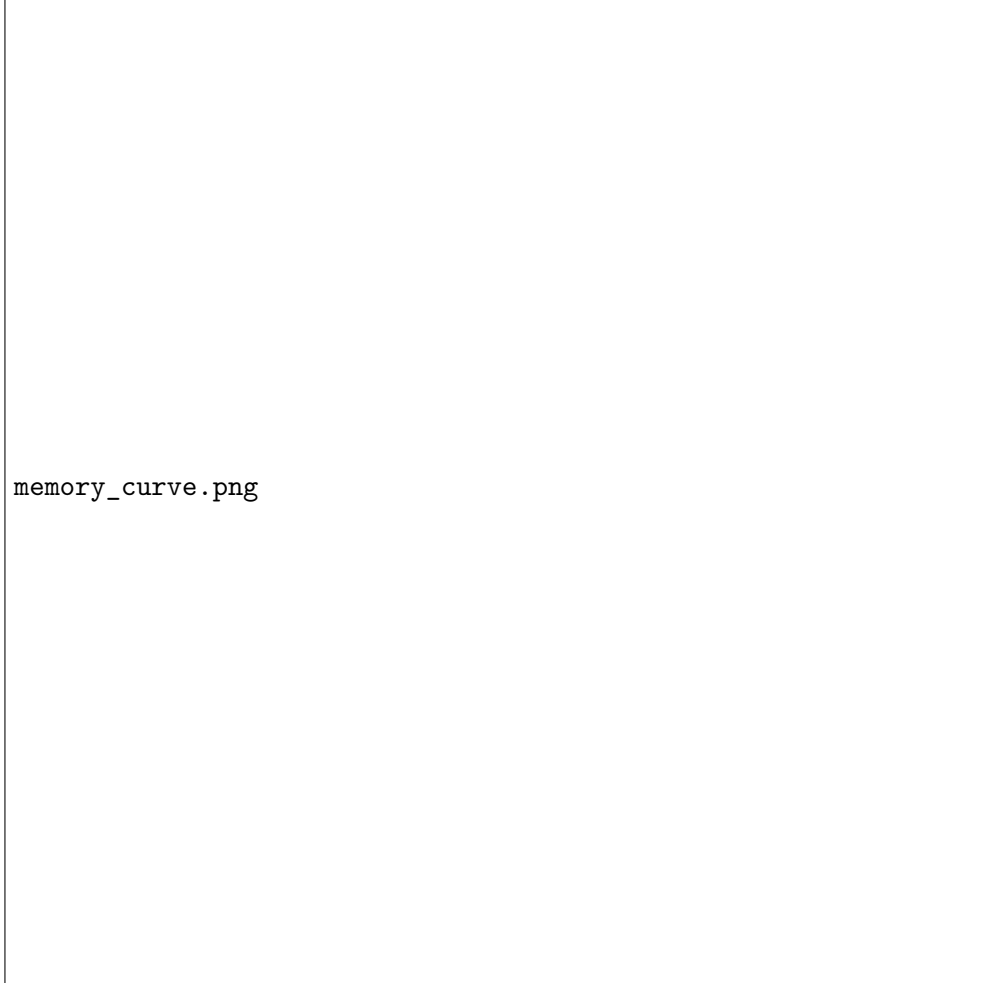


Figure 5: Memory augmentation benefit (Δ composite, MEMORY – MULTI) vs. cocoon store size. Two observed data points: non-significant at 217 cocoons ($\Delta = +0.004$, $p = 0.119$); large-effect significant at 951 cocoons ($\Delta = +0.031$, $p = 0.020$, $d = 0.80$). Curve is a $\sqrt{N}$ fit for illustration; systematic study is future work.

4. **Template-based agents in benchmark.** Agents use template-based reasoning in conditions where live LLM inference is not active simultaneously. Future work should conduct all evaluations with full LLM inference across all conditions.
5. **Depth–naturalness tradeoff (substantially resolved).** The April 2026 paper documented this as an open problem. In v8, CODETTE Turing naturalness improves from 0.245 to 0.820 without sacrificing composite score. The tradeoff appears tractable through deliberate response-structure engineering. Single benchmark suite validation on held-out problems is needed to confirm generalization.
6. **Strategy novelty measurement.** Novelty is measured relative to the existing strategy library rather than the broader literature. External validation is needed.
7. **Single model evaluation.** All benchmarks use Llama 3.1 8B. Generalization to other base models has not been tested.

8. **Theory scope.** The RC+ ξ framing is a conditional, assumption-dependent modeling lens. Establishing that real implementations satisfy the contraction premise (proposition 1) remains future work.
9. **Render integrity (new in v8).** The current render integrity check (word overlap $\geq 15\%$) is a weak proxy for semantic faithfulness. Embedding-space similarity between authored conclusion and rendered text would provide a stronger guarantee.

13 Conclusion and Future Work

We presented CODETTE, a cognitive architecture that models multi-perspective reasoning as a constrained dynamical system with embedded ethical constraints, meta-cognitive strategy evolution, intellectual integrity enforcement, and (in v8) clean render/cognition separation.

On the May 2026 benchmark (17 problems, 951 cocoons):

- Mean composite improvement: $+0.388$ ($0.357 \rightarrow 0.744$), $+108.8\%$, Cohen’s $d = 8.31$.
- Memory augmentation significant: $d = 0.80$, $p = 0.020$, resolving a prior null result at smaller scale.
- Depth–naturalness tradeoff resolved: CODETTE Turing naturalness $0.245 \rightarrow 0.820$ ($+235\%$); coherence $0.477 \rightarrow 0.700$.

Core contributions:

1. RC+ ξ formalism as a modeling framework for stabilizing forces in multi-perspective reasoning dynamics.
2. A working consumer-hardware implementation of 10-adapter multi-agent cognitive architecture with persistent memory at 951+ exchanges.
3. Phase 8 render/cognition separation: semantic authority resides in CognitionSubstrate; the LLM is a verbalization-only surface with AuthoredState integrity checking.
4. Evidence that memory benefit requires minimum scale: non-significant at 217 cocoons, large-effect significant at 951 cocoons.

Community reception. Following open-source release (Apache 2.0), the system received 1,800 views within one month via a public launch post [5]. Engagement included developers building competing cognitive runtime systems (whose independent identification of the model-coupling problem motivated Phase 8, section 5), non-expert users asking architectural questions about the reasoning pipeline vs. retrieval distinction, and French-speaking users expressing interest in offline-first reasoning for privacy-sensitive use cases. These interactions provided qualitative evidence that: (a) the system is accessible beyond the original author’s use cases, (b) the RAG vs. cognitive-pipeline distinction requires clearer documentation, and (c) privacy-preserving local deployment is a genuine user motivation distinct from pure research interest.

Future work: (1) human evaluation with inter-annotator agreement; (2) learning-curve analysis of memory benefit vs. cocoon count; (3) cross-model evaluation (Mistral, Gemma, Phi); (4) formal convergence proofs for RC+ ξ with explicit bounds; (5) held-out benchmark validation of Turing and coherence improvements; (6) render integrity strengthening via embedding-space faithfulness; (7) longitudinal study of strategy evolution; (8) structured user study incorporating community feedback from open-source deployment.

The system, benchmark suite, and supporting artifacts are archived on Zenodo (DOI: [10.5281/zenodo.19359663](https://doi.org/10.5281/zenodo.19359663)) and mirrored on GitHub/Hugging Face [? ? ? ?].

Author Contributions

J.H. (Jonathan Harrison): Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing — Original Draft Preparation, Writing

— Review & Editing, Visualization, Project Administration.

Funding

This research received no external funding.

A Supplementary Materials and Data Availability

To support reproducibility, this repository includes:

- Benchmark summary report (May 2026): `data/results/codette_benchmark_report.md`
- Benchmark results export (JSON): `data/results/codette_benchmark_results.json`
- Runtime benchmark exports: `data/results/codette_runtime_benchmark_20260526_*.md`
- Original raw outputs: `data/results/`
- Figures: `paper/figures/`

A.1 Overall Benchmark Results (May 2026, with standard deviations)

Table 5: Overall benchmark results by condition (mean $\pm$ std; $N = 17$ per condition; May 2026 run).

Cond.	Composite	Depth	Div.	Coh.	Eth.	Nov.	Grnd.	Turing
SINGLE	0.357 ± 0.055	0.369	0.324	0.381	0.088	0.439	0.395	0.431
MULTI	0.708 ± 0.037	0.854	0.946	0.668	0.390	0.706	0.612	0.582
MEMORY	0.739 ± 0.040	0.872	0.971	0.693	0.409	0.729	0.620	0.713
CODETTE	0.744 ± 0.036	0.863	0.966	0.700	0.387	0.701	0.641	0.820

A.2 Per-Category Results (May 2026)

Table 6: Composite scores by category (mean $\pm$ std; May 2026).

Category (N)	SINGLE	MULTI	MEMORY	CODETTE
Reasoning (3)	0.406 ± 0.032	0.719 ± 0.004	0.708 ± 0.012	0.721 ± 0.005
Ethics (3)	0.286 ± 0.044	0.674 ± 0.032	0.733 ± 0.024	0.723 ± 0.032
Creative (2)	0.392 ± 0.063	0.731 ± 0.025	0.772 ± 0.004	0.778 ± 0.050
Meta-cognitive (3)	0.355 ± 0.051	0.714 ± 0.051	0.763 ± 0.022	0.756 ± 0.048
Adversarial (3)	0.324 ± 0.020	0.706 ± 0.054	0.717 ± 0.068	0.749 ± 0.023
Turing (3)	0.388 ± 0.032	0.709 ± 0.041	0.749 ± 0.053	0.750 ± 0.048

A.3 Archived April 2026 Results (v7)

For longitudinal comparison, the April 2026 (v7) benchmark results (timestamp: 2026-04-08T20:59:44, 217 cocoons) are reproduced here:

Table 7: Overall results, April 2026 run (N=17 per condition).

Cond.	Comp.	Depth	Div.	Coh.	Eth.	Nov.	Grnd.	Turing
SINGLE	0.338	0.402	0.237	0.380	0.062	0.327	0.456	0.412
MULTI	0.632	0.755	0.969	0.503	0.336	0.786	0.604	0.180
MEMORY	0.636	0.770	0.956	0.500	0.340	0.736	0.599	0.291
CODETTE	0.652	0.855	0.994	0.477	0.391	0.693	0.622	0.245

Table 8: Evidence index: headline claims mapped to repository artifacts.

Claim	Where in paper	Primary evidence
Overall benchmark means (v8)	table 1	data/results/codette_benchmark_report.md
Memory augmentation significance	table 2	data/results/codette_benchmark_results.json
Turing improvement (+235%)	table 1, section 12	data/results/codette_benchmark_report.md
Coherence improvement	table 1	data/results/codette_benchmark_report.md
Cocoon count (951)	section 7	data/codette_memory.db
Longitudinal comparison (v7)	section A.3	data/results/codette_benchmark_report.md (April archive)

A.4 Evidence Index

References

- [1] AuraCoreCF. Comment on “I spent 3 years building a local AI that argues back, remembers everything, and won’t just tell you what you want to hear”. Reddit, r/THE_CODETTE_ROOM, May 2025. URL https://www.reddit.com/r/THE_CODETTE_ROOM/comments/1sx2gw2/. Independent practitioner evaluation; commenter is the developer of the Aura cognitive runtime system, who later provided a formal architectural review of CODETTE identifying the same model-coupling problem.
- [2] Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623, 2021.
- [3] Rishi Bommasani et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- [4] Aaron Grattafiori et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [5] Jonathan Harrison. I spent 3 years building a local AI that argues back, remembers everything, and won’t just tell you what you want to hear. here’s what i learned. Reddit, r/THE_CODETTE_ROOM, May 2025. URL https://www.reddit.com/r/THE_CODETTE_ROOM/comments/1sx2gw2/. 1,800 views; community launch post for the CODETTE open-source release.

- [6] G Robert J Hockey. Compensatory control in the regulation of human performance under stress and high workload: A cognitive-energetical framework. *Biological Psychology*, 45 (1-3):73–93, 1997.
- [7] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [8] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling (COLM)*, 2024. arXiv preprint arXiv:2311.12022.
- [9] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.
- [10] Peter Sterling. Allostasis: A model of predictive regulation. *Physiology & Behavior*, 106 (1):5–15, 2012.
- [11] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- [12] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.

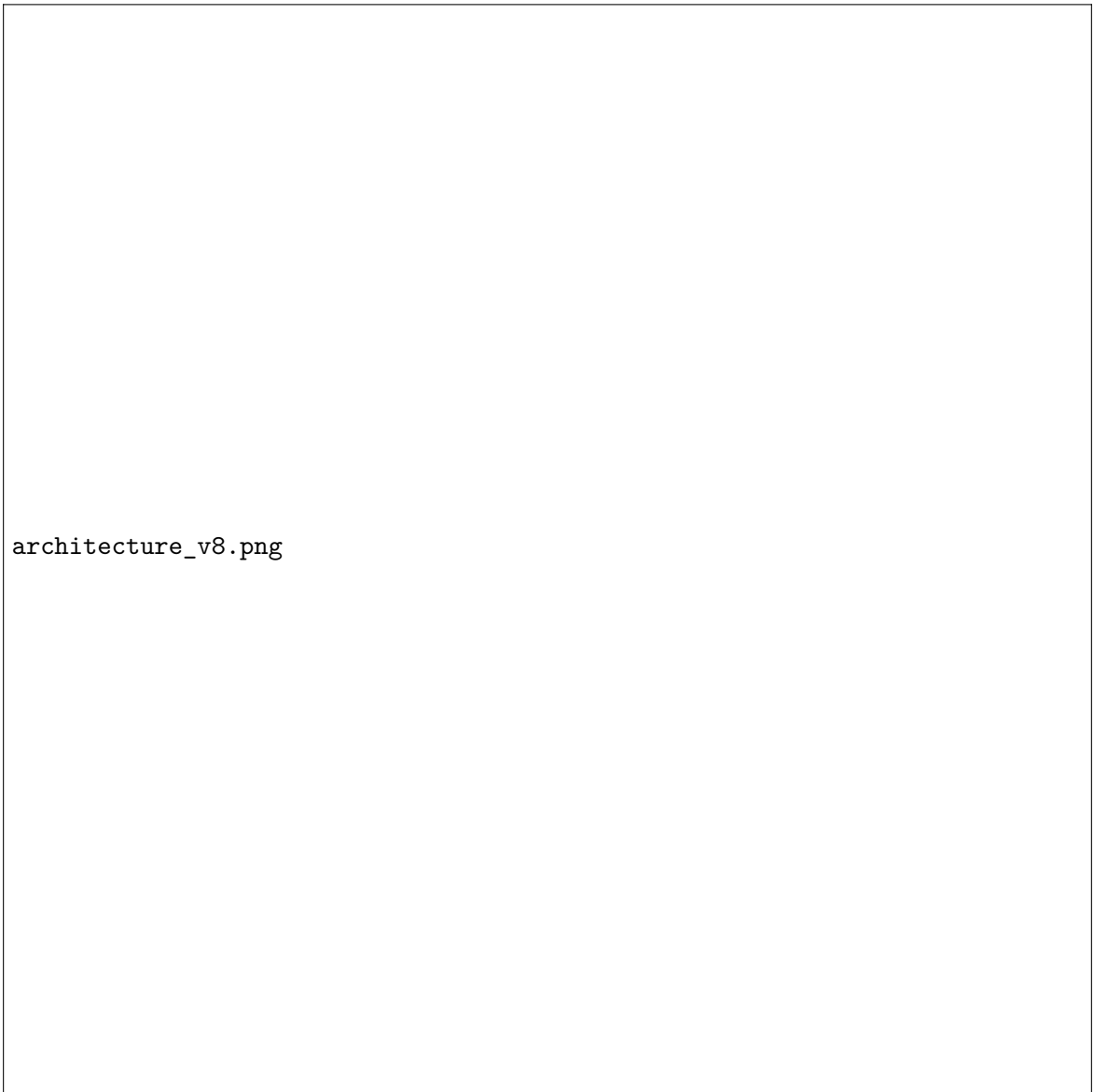
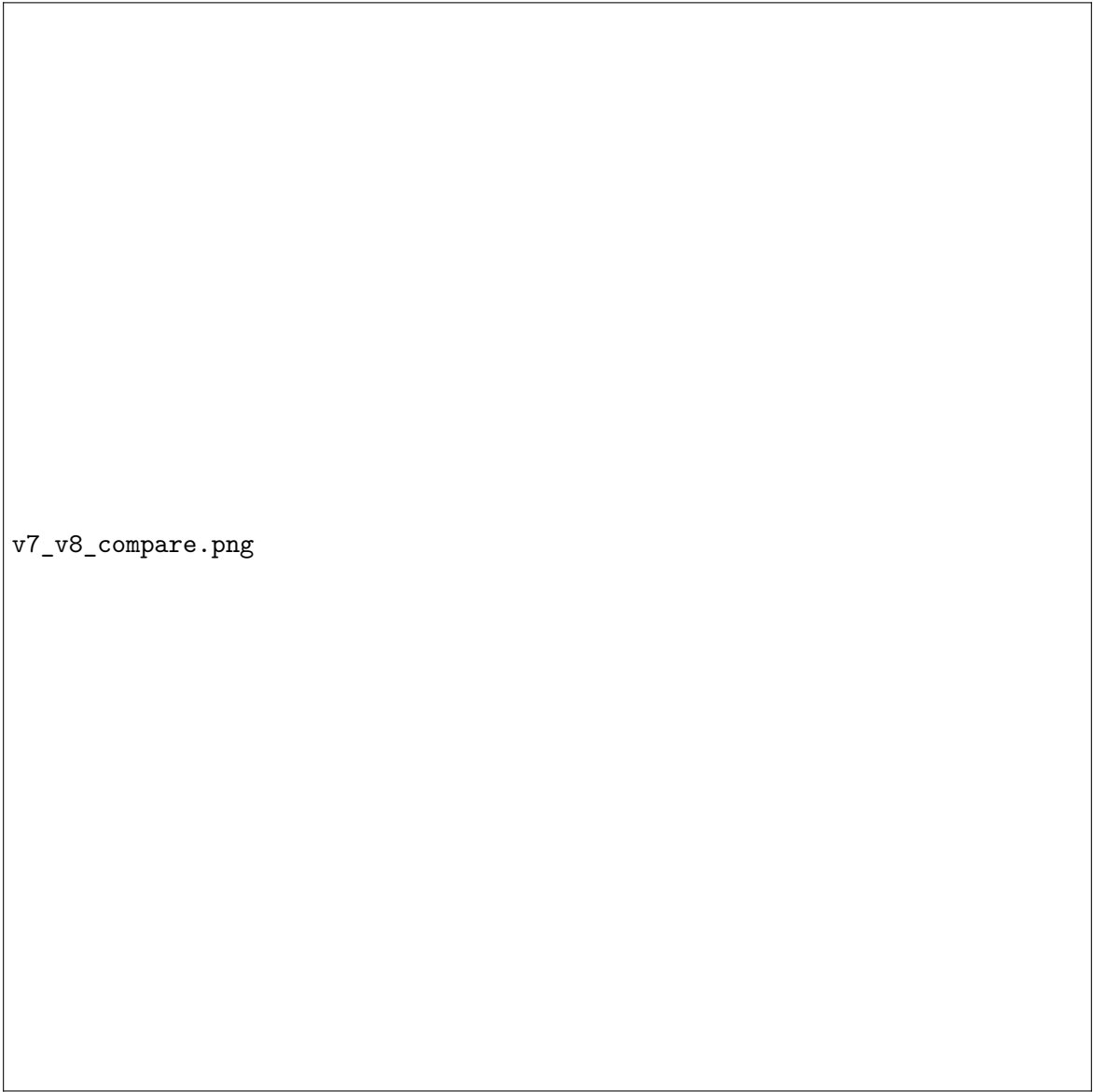


Figure 1: CODETTE v8 layered architecture. The top row shows the Phase 8 render/cognition pipeline (section 5): COGNITIONSUBSTRATE (zero LLM calls) produces an AUTHOREDSTATE, which RENDERLAYER expresses in natural language. The lower rows show the full system layers through which every query passes.



v7_v8_compare.png

Figure 2: Key metric improvements from April 2026 (v7, 217 cocoons) to May 2026 (v8, 951 cocoons). Turing naturalness improves by 235% ($0.245 \rightarrow 0.820$), resolving the depth–naturalness tradeoff documented in v7. Memory augmentation delta increases from 0.018 to 0.031 and reaches statistical significance ($p = 0.020$, $d = 0.80$).

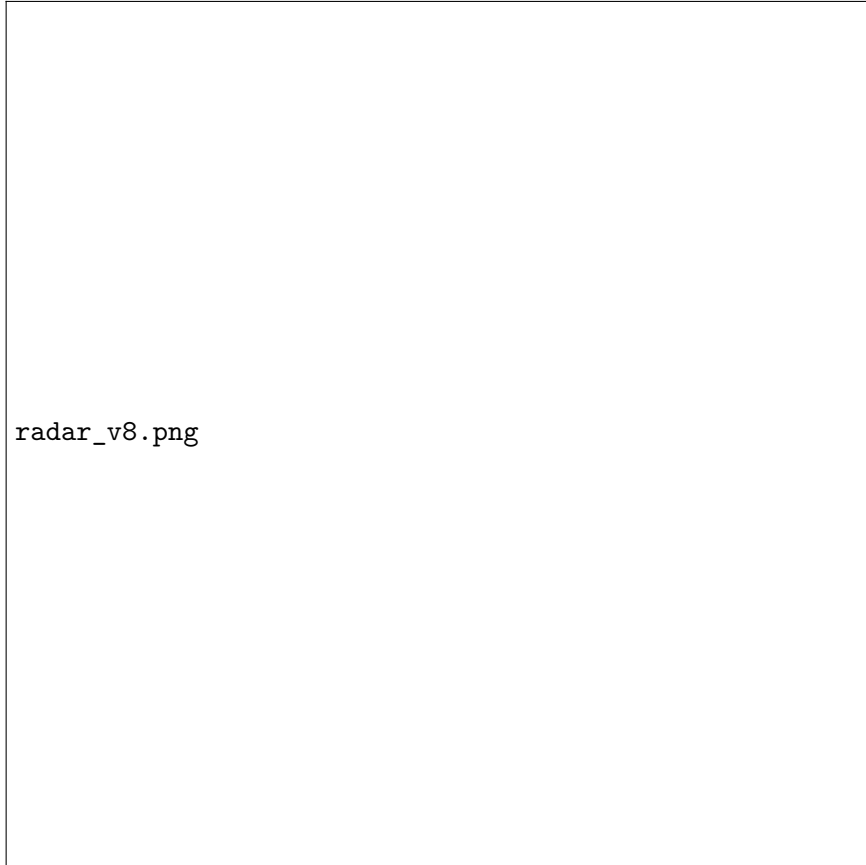


Figure 3: Radar plot of all seven scoring dimensions by condition (May 2026, $N = 17$ problems). CODETTE (red) leads on Turing naturalness and coherence; MULTI, MEMORY, and CODETTE cluster closely on perspective diversity and reasoning depth. All three multi-perspective conditions vastly outperform SINGLE (grey) across every dimension.



Figure 4: Runtime benchmark latency by category (May 26, 2026). Green bars indicate passing cases ($\text{score} \geq \text{target}$); grey bars indicate warnings. Valuation-reasoning queries (pre-computed graph traversal) complete in ~ 3.4 s; continuity-retention requires multi-turn context loading (~ 412 s). Mean latency: 115.2 s; pass rate: 60%.